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KONGU ENGINEERING COLLEGE
DEPARTMENT OF AI&DS

HCL PROJECT

Phishing Email Detection Using LSTM and Transformer-Based Deep Learning

Project Members:

ANTHONI MILTON R(24ADR009)
BHARATHI K(24ADR019)
BOMMI PRIYANKA (24ADR024)

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PROBLEM STATEMENT

- Phishing emails are increasing rapidly across organizations and individuals.
- Traditional spam filters fail to detect advanced phishing attacks effectively.
- Users often struggle to identify fake emails and malicious links.
- Manual email verification is time-consuming and prone to human error.
- An AI-based phishing detection system is needed for real-time cybersecurity protection.

LITERATURE SURVEY

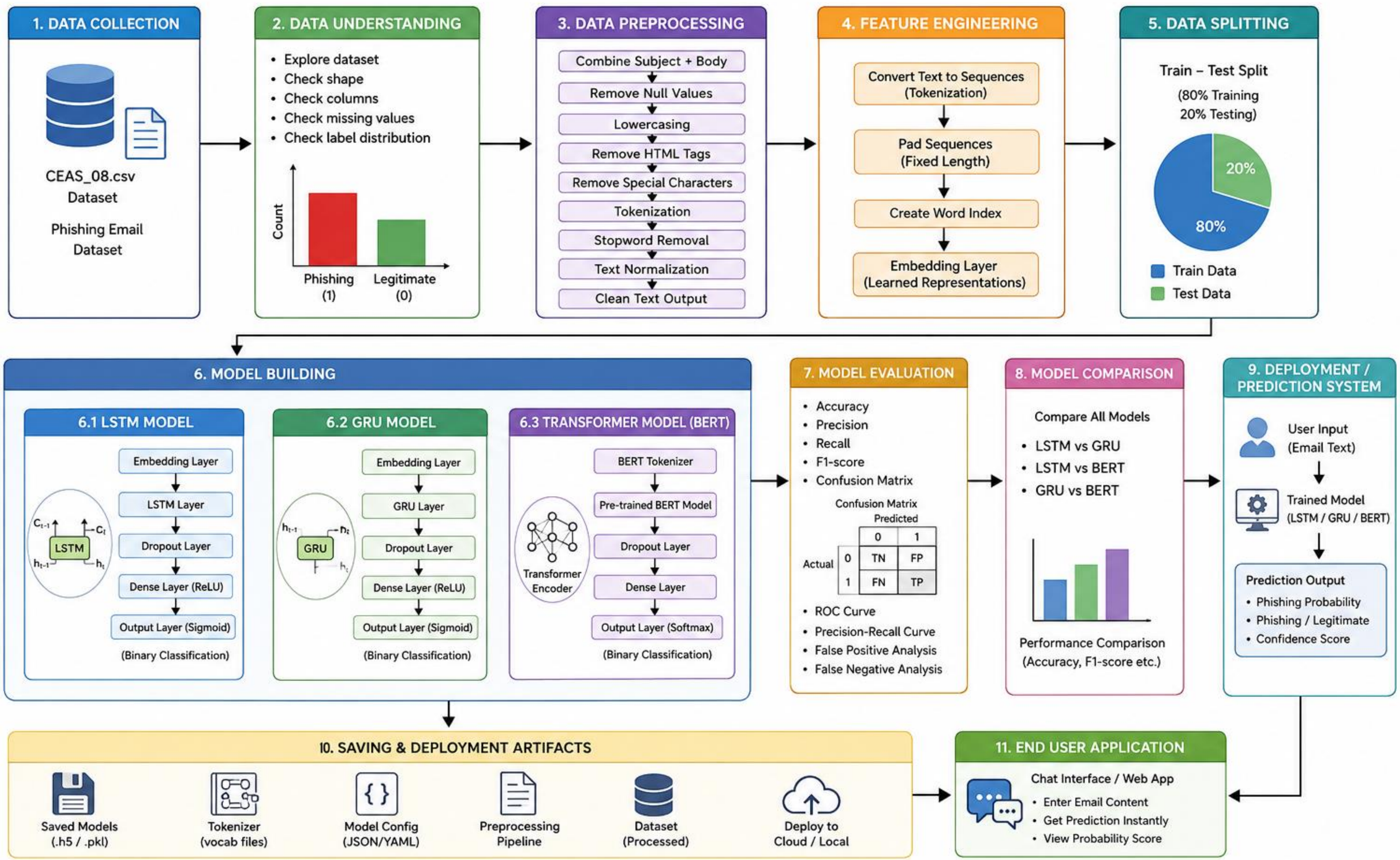
JOURNEL	YEAR	TITLE	METHODOLOGY	METRICS
PMC (National Institutes of Health)	2024	Advancing Phishing Email Detection: A Comparative Study of Deep Learning Models	LSTM, CNN, GRU, deep neural networks; comparison with traditional ML	DL models outperformed ML in accuracy; ML faster in computation time
Elsevier (Procedia Computer Science)	2021	Phishing Email Detection Using Natural Language Processing and Machine Learning	NLP, TF-IDF, ML classifiers (SVM, Naïve Bayes, Decision Tree)	Accuracy: 94.5%
JRASET Journal	2023	Phishing URL Detection Using LSTM-Based Ensemble Learning Approaches	LSTM, ensemble learning (bagging, stacking), URL features	Accuracy: >96.5%

OBJECTIVE

- To develop a deep learning-based email classification system for phishing detection.
- To provide instant phishing threat identification for enterprise email systems.
- To detect phishing emails using AI-based analysis of email content, subject line, and metadata.
- To improve access to cybersecurity protection against sophisticated phishing attacks.
- To create a simple and user-friendly email classification inference pipeline with phishing probability scoring.

WORKFLOW DIAGRAM – PHISHING EMAIL DETECTION SYSTEM

(LSTM, GRU & TRANSFORMER BASED APPROACH)



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DATASET DESCRIPITION

- The project uses the **CEAS 2008 Phishing Email Dataset**, which is widely used for cybersecurity and phishing email detection research.
- The dataset contains two categories of emails:
  - Phishing emails
  - Legitimate (safe) emails
- The dataset contains real-world phishing patterns such as fake account verification messages, urgent alerts, fraudulent links, and deceptive financial requests.
- The dataset is used to train and evaluate Deep Learning models like LSTM, GRU, and Transformer (BERT) for detecting phishing emails using Natural Language Processing techniques.

PROPOSED WORK

- Develop an AI-based phishing email detection system using Deep Learning and NLP techniques.
- Implement and train LSTM, GRU, and Transformer (BERT) models to classify emails as phishing or legitimate.
- Generate phishing probability scores and evaluate model performance using Accuracy, Precision, Recall, F1-score, ROC Curve, and Confusion Matrix.
- Build a real-time prediction system where users can input email content and receive phishing detection results.

MODULE DESCRIPTION

- **Data Collection Module**

  Collects phishing and legitimate emails from the CEAS 2008 dataset for training and testing the models.

- **Text Preprocessing Module**

  Performs text cleaning, tokenization, stopword removal, normalization, and sequence padding to prepare email data for Deep Learning models.

- **LSTM & GRU Model Module**

  Implements LSTM and GRU models to learn sequential patterns and detect suspicious phishing-related email content.

- **Transformer (BERT) Module**

  Uses Transformer-based BERT model to understand contextual semantics and deceptive language patterns in phishing emails.

- **Prediction Module**

  Generates phishing probability scores and classifies emails as phishing or legitimate based on model predictions.

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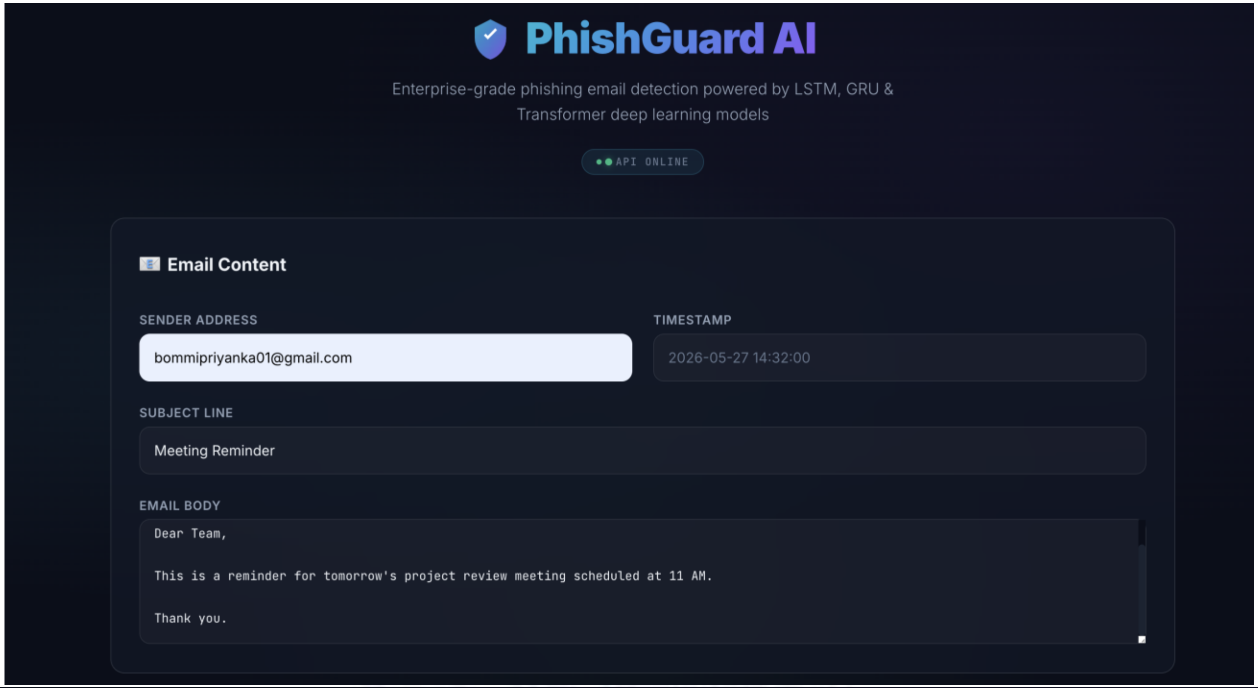
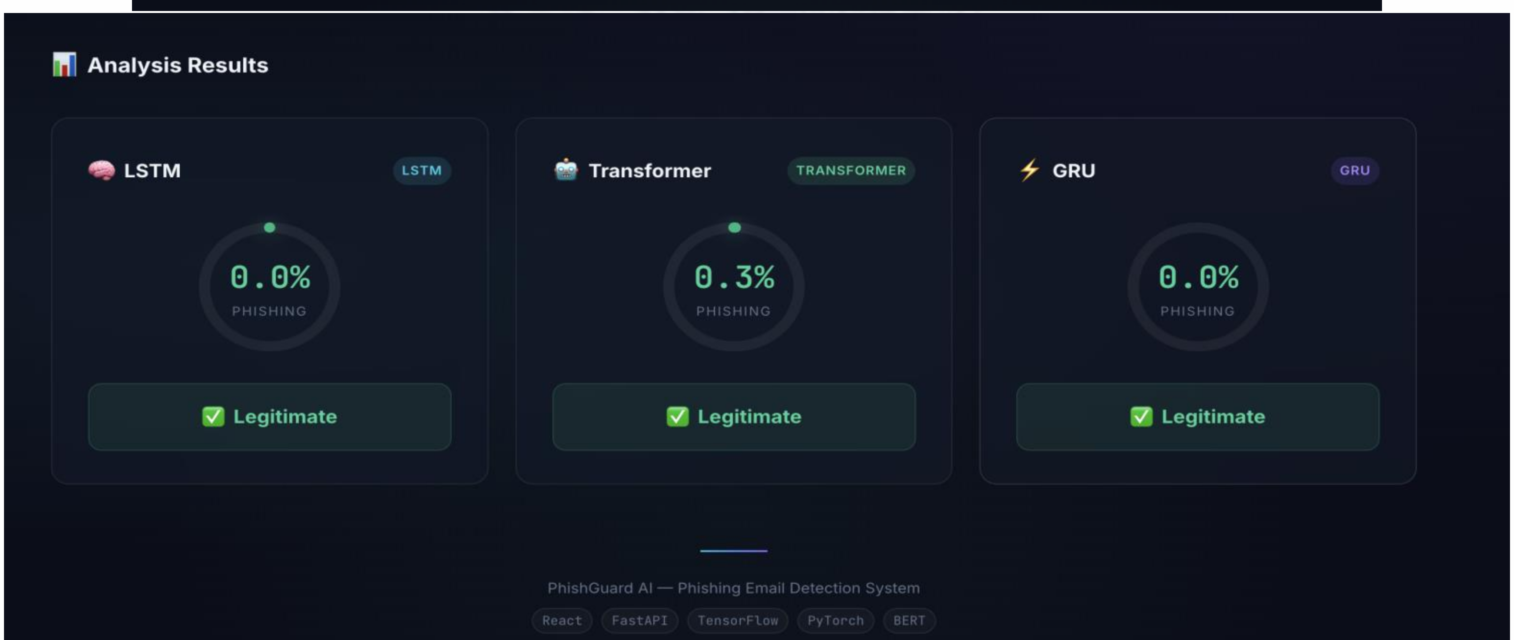
SCREENSHOTS





PERFORMANCE METRICS

- Accuracy → Measures overall prediction correctness.
- Precision → Measures how many predicted phishing emails are actually phishing.
- Recall → Measures how many phishing emails are correctly detected.
- F1-Score → Harmonic mean of Precision and Recall.
- ROC Curve → Evaluates phishing detection capability.
- Confusion Matrix → Displays TP, TN, FP, and FN values.

RESULTS ACHIEVED

- Implemented LSTM, GRU, and Transformer (BERT) models for email classification.
- Achieved effective phishing detection using Deep Learning and NLP techniques.
- Transformer model achieved better contextual understanding and higher prediction accuracy compared to LSTM and GRU.
- Generated phishing probability scores and real-time prediction results.
- Reduced false negatives and improved phishing email detection performance.

CONCLUSION

- The project successfully developed a Deep Learning-based phishing email detection system using NLP techniques.
- LSTM and GRU models effectively captured sequential email patterns, while the Transformer model provided better contextual understanding.
- The BERT model achieved better phishing detection performance compared to LSTM and GRU.
- The system can help organizations improve email security by automatically identifying phishing attacks.

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Thank You